Problem solving strategy-challenge for gifted students

Valentina Gogovska
Faculty of natural science, Institute of mathematics-Skopje, FYR of Macedonia

Abstract: Using problem solving strategy gifted students may experience the excitement of thinking deeply about mathematical ideas and discovering new concepts. They may learn to explore problems to find that the fun has only just begun when the original problem has been solved, because problem solution is often just the beginning of real mathematics.

The aims of this study are introducing some ideas and principles of problem solving strategy and showing practical tasks using mentioned methods by students individually.

At the end of this study the effects of problem solving will be analyzed.

Working with lot of students and seeing the difficulty that mathematical problem solving causes to my mathematical teaching students while they easily do other types of exercise, I realized that in teaching mathematics emphases should be put on the development of problem solving types of thinking.

Improvement of problem solving types of thinking doesn't mean that the students should learn as many mathematical tricks as possible to easy the problem solving of more difficult exercises. Unfortunately this can be observed in many schools. This method leads to exercises becoming a routine job and the process of problem solving does not appear in the solving of exercise at all. Why problem solving?

What a student does in response to the question that put understanding into action shows their level of understanding. The student might be able to solve an equation, but if there is no understanding of where the equation is coming from or where and how to use it, they may just be using a memorized skill that is going to be useful only for that type of equation, nothing more. Understanding a concept or a topic of study is being able to carry out a variety of actions or performances with the topic by the ways of critical thinking: explain, apply, transfer, generalize, represent in a new way, make analogies and metaphors, and so o. It is being able to take knowledge and use it in new ways (Perkins 1993). General guidelines lead teachers to recognize priorities when teaching for understanding: What shell we teach? What is worth understanding? How shall we teach for understanding? How can both students and teacher know what students understand and how can they develop deeper understanding? (Perkins1993) identified some of these priorities:(a) facilitating learning as a long-term, thinking-centered process;(b) providing rich ongoing assessment with supportive feedback and opportunities for reflection;(c) supporting with powerful representations;(d) recognizing flexible conceptions of what students can and cannot learn at certain ages; and (e) requiring the learner to carry facts and principles they acquire in one context into other contexts. Inducting talented students into mathematics as a system of thought that recognizes understanding in the nature of mathematics is embedded within these priorities. We believe that the existing breadth, dept and nature of these activities are necessary for gifted students.
However, after their finish students should be encouraged to discuss their work in group and hole class situations. Mathematics should not be a solitary activity, but a shared one, particularly in situations where there has been opportunity to show initiative. So

introducing problem solving strategy is interesting opportunity for better understanding from one way and for motivate them from other way.

A mathematical exercise can be called a problem if the solving of it is not routine job and if we use certain problem solving strategies and heuristic principles during the process. We can speak about problem solving if the mathematical theorems and procedures needed for the solution cannot be seen at the beginning, if there is more than one way to start and if there is more than one proper solution (open problems).

The process of problem solving according to Gyorgy Polya can be divided into four parts:

1. understanding the problem
2. making plans
3. executing the plan
4. reflection, looking back

If student want to experience these phase, the following things are needed:

-to understand the problem: knowing the mathematical concept present in the exercise base

-for plan-making: connection –system between the bases elements so as to find the connection between known and unknown problem solving strategies, heuristic principles

-executing the plan: demand that the procedures applied is mathematically correct.

-for reflection: recognizing the possible applications of exercise consequences picking out the special features of the exercise.

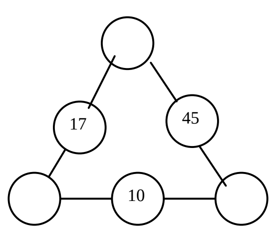
Among these phases, plan making is the most difficult one for students. Therefore main aim should be construction of the connection system between the concept needed for this phase, teaching problem solving strategies and heuristic principles.
For improvement of problem solving skills it is necessary that students see many problems during lessons, so the usual routine exercises should be changed to those problems that cannot be solved without the problem solving process.

Open problems

According to the characterization given in Japan, there are three kinds of open problems: open-ended problem, open-beginning problem, and both-ends-open problem (e.g. Hashimoto 2000, Nohda 2000). The concepts are explained below with some representative examples.

The following Number Triangle Problem is an example of an open-ended problem.

Example 1.The starting situation is given as follows:



(fig.1.)

There is a triangle where the corners are free and certain numbers are fixed on the sides (fig.1.)What numbers should be placed in the blank circles of the triangle so that the sum of the three numbers on each side is the same?

Since there are many solutions to this problem, we may continue e.g. with the following questions:

-Can you find another solution?

-How many different solutions could there be?

-Is it possible to use negative numbers in the circles?

-Can you find a solution where the triangle's side sum (i.e. the sum of the numbers on the same side) is 80?

-Try to pose a similar question. Can you answer it?

The following construction problem represents an open-beginning problem. Here the goal situation is exactly given: Find a box with the biggest volume.

Example 2.Construct an "open box" from an A4 sheet of paper by cutting off square pieces fro the corners and fixing the box with type. Compute the volume of the box constructed and compare your result with your peers' answers. How could you find (and construct) such a box that has the biggest volume?

The spiral of Ulams is an example of both-ends-open problems.

Example 3: Positive integers are written in the form of a spiral like

17	16	15	14	13	.
18	5	4	3	12	.
19	6	1	2	11	28
20	7	8	9	10	27
21	22	23	24	25	26

Find out a problem based on this number scheme, and solve it.

In the Example 3 a solver should first invent problem and formulate it (e.g.
Can you see any pattern in the diagonal number sequence 1,3,13 …beginning in the center? What is the n-th term of the sequence? e.c.t.)

A problem solving strategy is a previously learned method to solve a problem. For example Le Blanc (1997) has divided problem solving strategies into two groups: general and helpful strategies.

General strategies are:

1. Trial and error
2. A systematic list of the different possibilities
3. Simplifying the problem
4. Searching for the formula in the problem
5. Experimentation
6. Reasoning
7. Generalization
8. Working backwards

Helpful strategies are diagrams, tables, drawings, lists and equation.

These are indirect steps for implementing general strategies and are suitable for all general problem-solving strategies. Helpful strategies clarify and help to perceive problems that are posed.

Problem solving skills

This study used Moses (1982, 11.) for categorization of the following skills needed in problem solving:

Level1.Familiarization with the problem
*Motivation
Level2.Basic skills
*Mathematical skills and reading skills
Level 3.General cognitive skills
*Forming analogies
*Flexible thinking
*Visualization

A solving map

Usually the fastest and most talented pupils try to solve problems as an extra workload, without teacher's help. Most of them haven't the tools to solve verbal problems. The other pupils do routine tasks and they are not at all involved in problem solving. I think it is very important to teach some kind of tool to all pupils, especially for talented. The solving map is a method that helps students to understand and verbalize word problems, for example.
Instruction how to make a solving map

1. Familiarize yourself with the task and read it through as many times as you need to understand it!
2. Pick out information about task and write it down on a separate piece of paper at the beginning of the solving map.
3. Choose a solving strategy
4. Write down your own ideas on the solving map. Always make a drawing or diagram of the task if it is possible.
5. Evaluate and check your solution.
6. Improve your solution. If you find a lot of errors, make a new solution after the wrong solution.

Note! Don't delete the wrong solution! It is a part of the solving process too! Discuss an example of the problem.
Example of the solving map method

Word problem 1.

Some boys have a flashlight, which will last for only 12 minutes. They must get safely through a tunnel before the tunnel collapses. The boys have no escape route other than to go through the tunnel. Only two boys can go into the tunnel at the same time. It is too dangerous to go into the tunnel without a flashlight. It will take one minute for Oliver to go through the tunnel, for Tim it will take two, for Pete four and for Mark five minutes. Each couple must walk in the tunnel at the speed of the slower one. In which order must the boys walk through the tunnel so that they can go safely while the flashlight is on?

There are two parts to the solving map. Student makes own map on a separate sheet of paper. First student tries to find information about the problem.

A) Problem information
-A flashlight will last for 12 minutes and it must bi in the tunnel with boys
-Only two boys can go into the tunnel at the same time.
The boys' speeds in the tunnel are determined by the following times:
Oliver 1 min
Tim 2 min
Pete 4 min
Mark 5 min
-Each couple must walk in the tunnel at the speed of the slower one.

After that the student starts to solve the problem. Sentences and drawings are more than desirable!
B) Problem solution attempt:
1. Oliver takes Tim 2 min and comes back 1 min (sum.3 min)
2. Oliver takes Pete 4 min and comes back 1 min (sum.5 min)
3. Finally Oliver takes Mark 5 min (sum. 5 min)
Altogether 3min+5min+5min=13min
Wrong answer because the flashlight lasts only 12 minutes.
Now it is important that the pupils don't delete the wrong answer. The teacher must encourage student to continue and improve their answer.
New attempt:
Time must be saved: at least 1 min.
-Because each couple must walk in the tunnel at the speed of the slower one, it would be good to put the two slowest boys into the tunnel together!
-It would be useful to bring back the flashlight as fast as possible.
1. Oliver takes the flashlight and takes Tim 2min and Oliver comes back with the flashlight 1 min sum 3min.
2. Oliver gives flashlight to Pete who goes through the tunnel with Mark 5 min. Tim comes back with the flashlight 2 min.sum.7min
3. Lastly Oliver and Tim come through the tunnel together sum 2 min
Altogether 3min+7min+2min=12min!
What kind of problems would be good?
Changes in the word problem 1

First, we remove time limit of 12 min. Now everyone will certainly find some kind of answer.

Then we gradually increase the difficulty of the problem's next points so that the last point of the problem is closed.

a) In which order do you send the boys into the tunnel so that they can go quickly and safely?
b) Can you get the boys through the tunnel in 13 min?
c) Can you get the boys through the tunnel in 12 min? (It is possible!!)

The task should progress from an open to a closed problem so that no one hints a dead end right away. In this way we tempt students to familiarize themselves with the problem. When the first part of the problem has been solved, the threshold to try more demanding tasks is lowered because:
Success encourages the student to go on.
The problem has become familiar.
The students learn to work with the same problem longer than they are used to doing.

Example 1 :

Using Sofija Zermen identity:

$$a^4+4b^4=(a^2+2b^2+2ab)(a^2+2b^2-2ab), \tag{1}$$

talented students can solve tasks from this type:

a) Prove that for every $n>1$ number n^4+4 is compound.

b) Prove that exist infinitely many x who for every $n\in\mathbf{N}$, number $z=n^4+x$ is not prime.

v) Prove that number $2^{10}+5^{12}$ is compound.

g) Prove that number of type $4n^4+1$, $n\in\mathbf{N}$ is prime number if and only if $n=1$.

d**)** Prove that for every $n>1$ natural number n^4+4^n is compound. ♦

Example 2**.**a) It's easy to proved following task: *if M is point in the interior or on the side of equilateral triangular ABC and x,y,z are distances from M to the sides BC,CA,AB of triangle, ordered one after the other, than*

$$x+y+z=h, \tag{2}$$

where h is the height for the triangular.

b) Further, for students we give task from type:

Shepherd Pejo has only 3 pails, which are not dividing into grades for the quantity less than whole volume. First pail collect 12 *liter, second* pail 7 *liter, and the third pail collect* 5 *liter. The biggest pail is full with milk and Ilija wants to buy* 6 *liter milk. How can Pejo measure 6 liter milk for Ilija with given pails only?*

v) We make explanation for students discussing how previous task can be use for solving this or similar tasks. We explain that from previous tasks we directly conclude that when point M is moving in the interior of the equilateral triangular ABC or on his sides, than the sum of distances from the point to the sides of triangular is constant. Than if M is on the side BC, than $x=0$ and the opposite, if $x=0$, than M is on the side BC. Similar, $y=0$ if and only if M is on AC and $z=0$ if and only if M is on the side AB. Further, if point M is moving on the line parallel of side BC, than x is not changing, while y and z depend from the position of point M and (1) is satisfied.

Latter we discuss that

- Because sum $x+y+z$ is constant, we can look on this constant like on the hole quantity milk in the given tri pails, point moving on the line parallel of some triangular side we can consider like pour from one pail in the other, meaning that if with x we mark the quantity of milk in the first , than moving of M on the line parallel with side BC we count like pouring from the second to the third pail and opposite, and
- every side from the equilateral triangular ABC with height 12 we separate on 12 equal pieces and from the points of intersection we drew lines parallel with triangular sides. Then we get triangular integer bars, which top point are ordered triples integers, coordinates, which are distances of the top to the sides BC, CA, AB , one after other. So, for example, point A has coordinate (12,0,0), B (0,12,0) and C (0,12,0) est.

g) we make explanation for students that with previous action they can solve tasks in the next two cases which are:

- if volume of the bigger sad is bigger than the sum of the volumes of the others two pails, than permission area is parallelogram which top is in the interior of the equilateral triangular and
- if volume of the bigger pail is lower of the sum of volumes of the others two pails, than permissible region is pentagon with two top points on the side BC . ♦

Problem solving makes challenging situation and open approach for gifted students: they can always go further, go beyond situations, ask new questions, initiate their own investigations, and be more creative in their mathematical work.

At the same time this gives us as teachers a chance to understand better what it is that makes mathematical talent appear and grow and thus leads to the creation of more efficient didactical tools that would help to keep their interest in learning mathematics.

The trials for problem solving for lower-achieving students are blind, unmotivated, and unsystematic. On the contrary gifted students have an organized plan of searching. Talented students switch easily from one mental operation to another, they have great flexibility and mobility in their mental processes in solving mathematical problems, and it is therefore easy for them to reconstruct established thought patterns. For average students it is much harder to switch to a new method of problem-solving. The talented students thoroughly investigate the problem, which may suggest that they enjoy working with mathematics. The emotional factor is seen in that they often try to solve the problem in a more simple way or improve the solution and they show satisfaction when the solution was economical, rational and elegant.

This kind of activity shows how the engagement of talented students in PROBLEM SOLVING process makes it more meaningful for them.

Literature

1. ProMath Jena 2003 : Problem solving in Mathematics Education, 2004 edited by Hartmut Rehlich and Bernd Zimmermann
2. Malceski, R; Gogovska,V. Integration of mathematics curriculum, a Challenge of the Contemporary Mathematics
3. Moses, B.1982: Individual differences in problem solving. Arithmetic Teacher 30,4,10-14
4. Nohda, N.2000: Teaching by Open-Approach Method in Japanese Mathematics Classroom. In Proceedings of the PME-24 Conference
5. Malceski, R.: And this is easy (algorithm for case solving), Numerus, 2004/05
6. Polya, G.: how to solve it? Second edition Princeton, New Jersey: Princeton University Press
7. Le Blanc, J.F.1977: You can teach problem solving. Arithmetic Teacher .Vol25